Everything Has a Slope

Teaching mathematics by modelling the world around us

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1. Where my teaching starts

Most students meet slope as a formula on a whiteboard. They learn to subtract two y-values, subtract two x-values, and divide. Many of them get the right answer. Far fewer could walk outside, look at two ramps, and tell me which one is steeper and why.

I would rather they meet slope on a ramp.

I want students to see mathematics as a way to investigate their world. I begin with situations they can explore, let them make predictions and learn from what happens, and help them develop the mathematical tools to explain what they discover.

Behind this are a few beliefs I hold about learning. Students understand an idea more deeply when they have tried something, watched it fail, and worked out why. Mathematics is worth learning because it helps people make decisions about real things, and students should feel that. And a wrong answer is useful to me as a teacher, because it shows me exactly how a student is thinking.

Mathematical modelling is how I put these beliefs into practice. Students take a real situation, decide what matters, describe it with mathematics, make a prediction, and check that prediction against reality. When it doesn't match, they revise.

Going outdoors is not the method itself. It is the setting that makes the method tangible. A ramp doesn't come with labelled axes. Nobody tells you which length to measure. Students have to make those decisions, and making them is where the mathematics starts.

> **The world gives students questions. Mathematics gives them tools to investigate. Modelling teaches them to judge whether their answers make sense.**

I chose slope for this case study because it is everywhere: ramps, stairs, walls, roofs, slides, hills, and every graph students will ever draw. It is also one idea that runs all the way from junior secondary to calculus, even though the school curriculum rarely presents it that way.

2. Two ways to teach slope

A typical lesson on slope goes like this. The teacher writes the definition:

$$\text{gradient} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Then comes a worked example with two points on a grid, followed by an exercise. Students practise until the procedure is reliable.

This builds fluency, and fluency matters. But I often see students who can calculate a gradient perfectly and still can't tell me:

- why we divide rise by run instead of subtracting one from the other
- what the number actually means for the ramp in front of them
- whether their answer is reasonable

	Traditional lesson	Exploration lesson
Starts with	The formula	A ramp and a question
Students' first job	Follow the worked example	Make a guess, then find a way to check it
Why divide?	Because the formula says so	Because subtracting gave a wrong answer
Data	Given, clean, labelled	Collected, messy, argued over
Ends with	Exercises completed	A decision: does this ramp pass or fail?

There is a second problem, and it is less obvious. In Hong Kong, students meet slope in separate pieces across several years:

- the gradient of a straight line in junior secondary
- the angle of inclination in trigonometry
- positive, negative, zero and undefined gradients as rules to memorise
- the gradient of a curve and differentiation in the Extended Module, years later
- the slope of a distance–time graph in Physics

Each appears as its own chapter, and students rarely realise they are one idea. In the lesson below, students walk from a ramp to a staircase to a wall to a slide in one investigation. By the end they have met almost every one of those pieces, and they have seen how the pieces connect.

3. Part A: Could a wheelchair user get up this ramp?

The setting

Two ramps near the school or in a nearby park. Ramp A is long and rises quite high. Ramp B is short and rises only a little. Students need a tape measure, a 1 m spirit level, a ruler, and a phone with a free clinometer app.

The driving question is:

> **Could someone in a wheelchair get up this ramp on their own?**

I chose this question because students can't answer it by looking. They first have to work out what "steep" means and how to measure it. The question also has a real standard to check against. Hong Kong's barrier-free design guidance sets 1:12 as the steepest gradient allowed for a ramp. That means the investigation ends with a decision, not just a number.

The lesson moves through seven stages. For each one I've noted what I ask, what students usually do, the mistake I expect, and how I respond.

Stage 1: Notice

> **"Which ramp is steeper? Point to one."**

Almost everyone points to Ramp A. It is taller and longer, and it looks more impressive.

> **"How could you prove it to someone who wasn't here?"**

This is the real start of the lesson. Pointing is no longer enough. Students need a method that someone else could repeat.

The mistake I expect: taller means steeper, or longer means steeper.

How I respond: I don't correct it. I let the disagreement build, because the measurements will settle it later, and students remember the moment their own data proves them wrong.

> **From my lesson:** *[Add what your two students guessed first and what they said.]*

Stage 2: Decide what "steep" means

> **"What would you measure to prove it?"**

Students suggest the height, the length, or the angle. Each suggestion is reasonable, so I test each one with a counterexample.

- **If they say height:** "A ramp rises 1 metre, but it stretches the length of a football pitch. Is it steep?"
- **If they say length:** "This ramp is only 50 cm long. Is it steep or gentle?" (It could be either.)
- **If they say angle:** "Good. How would you measure the angle of this ramp with what's in your bag?"

Height alone doesn't work, and neither does length alone. Students start to realise that steepness depends on how the height compares with the length. That is the idea of a ratio, although nobody has used the word yet.

Stage 3: Measure

Students plan their own measurements. Usually they measure the total height of the top of the ramp and the length of the ramp.

Some will measure along the sloping surface. Others will measure along the ground underneath. I let both happen and come back to it in Stage 6.

If nobody thinks of it, I show them the spirit-level trick. Hold the 1 m level horizontally with one end touching the ramp. Measure the gap between the other end and the ramp surface. That gap is how much the ramp rises over one metre.

> **"If your partner measured it again, would they get the same number?"**

Students compare readings and find small differences. Where did you start the tape? Was it sagging? Did you mix centimetres and metres? These are not distractions from the mathematics. They are the start of judging evidence.

Stage 4: Make the ramps comparable

Students now have numbers, but the ramps are different sizes, so the raw numbers can't be compared directly.

	Horizontal run	Rise
Ramp A	6.0 m	45 cm
Ramp B	1.2 m	15 cm

> **"If each ramp went exactly 1 metre across, how far up would it go?"**

Ramp A rises 45 cm over 6 m, which is 7.5 cm per metre. Ramp B rises 15 cm over 1.2 m, which is 12.5 cm per metre.

The short, low ramp is steeper. This is the moment students usually react, because it contradicts what they pointed to in Stage 1.

Students have just invented the gradient: rise per unit of run. Only now do I give it a name.

The mistake I expect: some students subtract instead of divide, for example run minus rise.

How I respond:

> **"Imagine Ramp B made twice as big in every direction. Is it steeper?"**

The bigger ramp runs 2.4 m and rises 30 cm. It looks exactly as steep, and the ratio stays the same. But the difference has doubled from 1.05 m to 2.10 m. So subtraction can't be measuring steepness. Students reach this conclusion themselves, which is why it stays with them.

> **From my lesson:** *[Add the moment your students realised they needed to compare, not just measure.]*

Stage 5: Predict and test

Now I bring in the standard.

> **"Hong Kong says a ramp can be no steeper than 1:12. Do our ramps pass?"**

1:12 means rising 1 unit for every 12 across, a gradient of about 0.083. Ramp A (0.075) passes. Ramp B (0.125) fails.